AP PRECALCULUS

Unit 1 • Lesson 1.8 practice • Answer key

Lesson 1.8 • Rational Functions and Zeros

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

What are the zeros of $r(x) = \dfrac{x^2 - 9}{x + 2}$?

(A) -2
(B) 3 and -3
(C) 9
(D) 3 only

Solution

Answer: (B). The numerator is 0 at $x = \pm 3$, and neither value makes the denominator 0.

Question 2

What are the zeros of $r(x) = \dfrac{x^2 - 4x}{x - 4}$?

(A) 0 and 4
(B) 4 only
(C) 0 only
(D) There are none.

Solution

Answer: (C). $x = 4$ is not in the domain of r, so only $x = 0$ is a zero.

Question 3

What are the zeros of $r(x) = \dfrac{(x + 1)(x - 5)}{(x - 5)(x + 3)}$?

(A) -1 only
(B) -1 and 5
(C) 5 only
(D) -3 and -1

Solution

Answer: (A). $x = 5$ is excluded from the domain (there is a hole), and $x = -3$ is a zero of the denominator.

Question 4

What is the solution set of $\dfrac{x + 2}{x - 1} < 0$?

(A) $(-\infty, -2)$
(B) $(1, \infty)$
(C) $(-\infty, -2) \cup (1, \infty)$
(D) $(-2, 1)$

Solution

Answer: (D). The sign changes at -2 and 1. Test $x = 0$: $\frac{2}{-1} < 0$. Outside the interval the quotient is positive.

Question 5 (Free response)

Let $r(x) = \dfrac{x^2 - x - 2}{x^2 - 9}$.

(a) Factor the numerator and denominator, and find the zeros of r.
(b) Make a sign chart for r and solve $r(x) > 0$.

Solution

(a) $r(x) = \dfrac{(x-2)(x+1)}{(x-3)(x+3)}$. No factor cancels. Zeros: $x = 2$ and $x = -1$.

(b) The sign can change at -3, -1, 2, 3. Test values: $x = -4$: $\frac{(-6)(-3)}{(-7)(-1)} > 0$; $x = -2$: $\frac{(-4)(-1)}{(-5)(1)} < 0$; $x = 0$: $\frac{(-2)(1)}{(-3)(3)} > 0$; $x = 2.5$: $\frac{(0.5)(3.5)}{(-0.5)(5.5)} < 0$; $x = 4$: $\frac{(2)(5)}{(1)(7)} > 0$.
So $r(x) > 0$ on $(-\infty, -3) \cup (-1, 2) \cup (3, \infty)$.